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Three-dimensional reconstruction Based on Tri-frequency Heterodyne Principle

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Abstract

The traditional 3d reconstruction method of stereo vision is based on image feature point extraction, but it is very inconvenient to extract feature points in more complex cases, and the reconstructed 3D information is not comprehensive. In this paper, a 3d reconstruction method based on three-frequency heterodyne four-step phase shift method is proposed. System parameters is obtained by binocular calibration algorithm first, and then using digital grating projector to the object under test of three groups of different frequency phase shift fringe and collecting images, using the four step phase-shift and three phase frequency heterodyne method to get the main value and get the absolute phase of continuous distribution, combining with the polar correction and phase matching reconstruction of 3 d point cloud data of the object to be tested, finally complete 3 d measurement in 3 d software. The experimental results show that the system can realize 3d dimension measurement of heterogeneous castings with high detection efficiency.

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1. Introduction

Three-dimensional reconstruction is a hot research topic in the field of science and technology at present. It can recover the three-dimensional coordinates of objects and shape information according to the picture information,

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and is widely used in agriculture¹, medical field², autonomous driving field³, cultural relic protection⁴, traffic environmental flow monitoring⁵ and other aspects.

Many experts and scholars at home and abroad have done a lot of research work in the field of THREE-DIMENSIONAL optical measurement. For example, Ai Jia⁶ based on the principle of three-frequency heterodyne and monocular vision, used the rapid projection of projection module and the synchronous acquisition of CCD camera to build a set of measurement system for the surface topography of objects under the micro field of view (8mm × 6mm), and successfully reconstructed the surface 3D topography of objects represented by COINS in the depth range of 100 m. Guo Jin⁷ developed a set of THREE-DIMENSIONAL measurement system based on the principle of multi-frequency heterodyne to solve the problem of large phase error in the measurement of object edge by combining gray code with phase shift technology. The system was able to complete the complex surface type measurement and obtain high-precision point cloud data.

In this paper, the phase grating method⁸ is used to expand the phase of the projected fringe image by using the three-frequency heterodyne principle, which is suitable for static object morphology measurement. In this paper, the measurement system using two sets of CCD camera and an imaging lens binocular vision system, projecting DLP chip development and the projection of the component standard sine fringes to the measured object surface, software synchronization trigger controls the receiving of the stripe image CCD camera, after three frequency heterodyne algorithm for image processing, realize the measured parts of 3 d reconstruction and measurement of surface topography.

2. System scheme design

The hardware system consists of four parts: a computer, a DLP digital projector (grating projection device), two CCD cameras (image acquisition device) and a mechanical fixed device, as shown in the figure below.

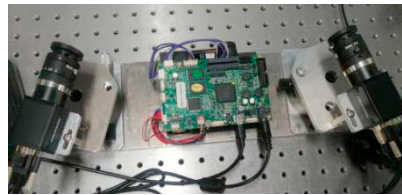


Fig. 1. Mechanical fixtures and equipment display.

3. Camera calibration

The main difference between binocular camera calibration¹⁰⁻¹² and monocular camera calibration is that in addition to obtaining the internal and external parameters of each camera, the relative relationship of rotation and translation between the right camera and the left camera in the world coordinate system needs to be determined.

According to the mathematical model, it can be obtained that:

$$\begin{bmatrix} X_{cl} \\ Y_{cl} \\ Z_{cl} \end{bmatrix} = R_l \begin{bmatrix} X_w \\ Y_w \\ Z_w \end{bmatrix} + T \quad \begin{bmatrix} X_{cr} \\ Y_{cr} \\ Z_{cr} \end{bmatrix} = R_r \begin{bmatrix} X_w \\ Y_w \\ Z_w \end{bmatrix} + T_{rl} \quad (1)$$

Simultaneous solution can be obtained as follows:

$$\begin{bmatrix} X_{cl} \\ Y_{cl} \\ Z_{cl} \end{bmatrix} = R_l R_r^{-1} \begin{bmatrix} X_{cr} \\ Y_{cr} \\ Z_{cr} \end{bmatrix} + T_l - R_l R_r^{-1} T_r \quad (2)$$

By solving the above equation, the final rotation and translation matrix expression can be obtained, as follows:

$$\begin{cases} R = R_l R_r^{-1} \\ T = T_l - R_l R_r^{-1} T_r \end{cases} \quad (3)$$

4. Phase grating method and its three-frequency heterodyne principle

4.1. Principle of phase displacement

Phase grating method belongs to phase measurement profilometry and is widely used. The basic principle is that when the DLP projector is used to project the coded grating to the surface of the object under test, the projected light will be modulated by the surface morphology of the object to form the deformed grating, and the image of the deformed grating will be received by the CCD camera. If the intensity of projected light meets the standard sinusoidal distribution, the light intensity distribution function of the camera acquired deformed grating is

$$I(x, y) = A(x, y) + B(x, y) \times \cos[2\pi\tau x + \Delta\varphi(x, y)] \quad (4)$$

In formula (4), $I(x, y)$ and $A(x, y)$ are the light intensity and the object background light intensity at different locations of the image respectively, $B(x, y)$ is the modulation amplitude, the frequency of the stripes, and the phase function refers to the phase change of the transformation grating, which contains the 3D information of the object surface. By demodulating the phase function, the three-dimensional topography of the object can be measured. At present, the commonly used phase-shifting methods include phase-shifting method and frequency-domain method. In this paper, the more mature N-step phase-shifting method is used. The n-step phase shift method is mainly used to process the deformed grating images in the spatial domain. The projected sinusoidal fringes move once every fixed grating period $1/N$, and the corresponding light intensity function $I_n(x, y)$ is generated at the same time. The phase of the fringes actually moves $2\pi/N$. Considering that the four-step phase shift method can eliminate the background term and effectively suppress the nonlinear response of the measurement system, and the number of projected fringe amplitude is not the maximum, this paper adopts the four-step phase shift algorithm⁹, with each phase shift being $\pi/2$, and the intensity function of the four deformation fringes obtained is

$$\begin{cases} I_1(x, y) = A(x, y) + B(x, y) \times \cos[2\pi\tau x + \Delta\varphi(x, y)] \\ I_2(x, y) = A(x, y) - B(x, y) \times \sin[2\pi\tau x + \Delta\varphi(x, y)] \\ I_3(x, y) = A(x, y) - B(x, y) \times \cos[2\pi\tau x + \Delta\varphi(x, y)] \\ I_4(x, y) = A(x, y) + B(x, y) \times \sin[2\pi\tau x + \Delta\varphi(x, y)] \end{cases} \quad (5)$$

The phase distribution function of the object surface can be calculated by using Equation (6)

$$\Delta\varphi(x, y) = \arctan\left[\frac{I_4(x, y) - I_2(x, y)}{I_1(x, y) - I_3(x, y)}\right] \quad (6)$$

4.2. Three frequency heterodyne method

The phase value calculated by the four-step¹³⁻¹⁴ phase shift algorithm is truncated between $\pm\pi$ by the arctangent function, resulting in the non-unique relative phase value of each pixel in the whole measurement range. Therefore, it is necessary to expand the phase value of the spatial point to obtain the continuous phase distribution, which is realized by the three-frequency heterodyne method in this paper.

The three-frequency heterodyne method combines the heterodyne principle and the three-frequency unrolling method, and the phase distribution functions of $\phi_1(x,y)$, $\phi_2(x,y)$ and $\phi_3(x,y)$ of the frequencies λ_1 , λ_2 and λ_3 are projected onto the surface of the object under test, respectively, and the corresponding fringe images are taken synchronously in turn. Three truncated phase graphs are obtained by the four-step phase shift algorithm, and the phase of each point is expanded along the time axis independently by using the third heterodyne. In fact, heterodyne is the difference of the phases of the corresponding pixel points, from which the absolute phase values of different parts of the object can be obtained, so as to avoid the propagation of errors in principle. In order to accurately expand the phase in the whole measurement range, appropriate λ_1 , λ_2 and λ_3 value should be selected to make the heterodyne superposition for three times and finally obtain the phase diagram with fringe period of 1, as shown in Fig. 2 below:

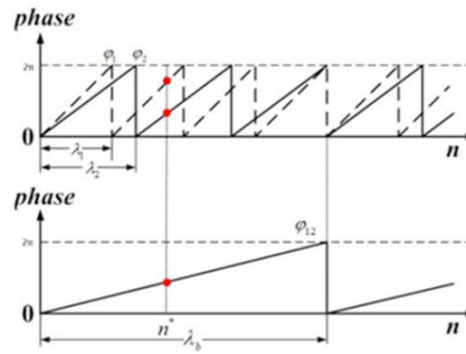


Fig. 2. Schematic diagram of heterodyne principle

$$\lambda_b = \frac{\lambda_1 \lambda_2}{\lambda_1 - \lambda_2} \quad (7)$$

The heterodyne principle can be used to expand the relative phase value of spatial points. In order to unambiguously expand the phase value in the full field, the appropriate λ_1 and λ_2 value must be selected, such that $\lambda_b = 1$. As shown in FIG. 6, the following formula $\Phi(\lambda_i)$ can be used for phase unrolling:

$$\Phi(\lambda_i) = \varphi(\lambda_i) + 2\pi \times \text{round} \left[\frac{\Phi(\lambda_{i-1}) \times (\lambda_i / \lambda_{i-1}) - \varphi(\lambda_i)}{2\pi} \right] \quad (8)$$

Where, $\Phi(\lambda_i)$ is the unwrapping phase of the phase-shifted fringe of frequency λ_i , and $\varphi(\lambda_i)$ is the wrapping phase of the phase-shifted fringe of frequency λ_i .

Respectively in this paper, we use the projection frequency¹⁵ $\lambda_1 = 1/70$, $\lambda_2 = 1/64$ and $\lambda_3 = 1/59$ three raster image, the phase shift algorithm to get the corresponding Δ phase distribution function $\Delta\phi_1(x, y)$, $\Delta\phi_2(x, y)$ and $\Delta\phi_3(x, y)$. Respectively using heterodyne principle of superposition $\Delta\phi_1(x, y)$, $\Delta\phi_2(x, y)$ and $\Delta\phi_3(x, y)$, get the frequency for

$\lambda_{12} = 1/6$ and $\lambda_{23} = 1/5$ phase Φ_{12} and Φ_{23} , then Φ_{12} and Φ_{23} overlay, get in the entire measuring range is only one cycle phase Φ_{123} , its frequency $\lambda_{123} = 1$, the principle is shown in Fig. 3.

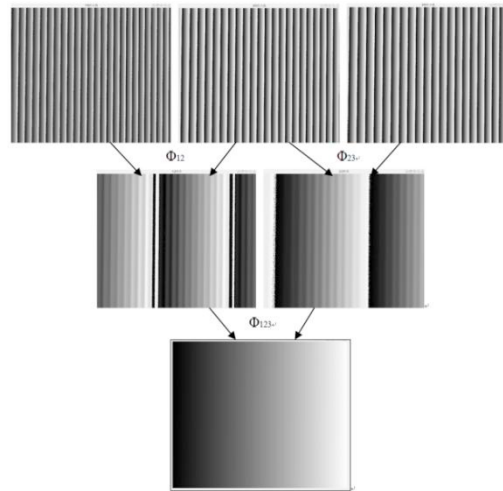


Fig. 3. Effect diagram of heterodyne principle

5. Matching method based on phase constraint

For traditional binocular stereo vision, the image matching is usually analyzed and processed in the natural state. However, when the surface of the measured object is too complex or too simple, or when there is no significant difference in the reflection of the projected light on the surface of the measured object, the grayscale difference between the points in the generated image will be very weak, so it is difficult to match it with the traditional matching method¹⁶⁻¹⁸.

This paper presents a method to match image points with phase constraints¹⁹. The measurement system built in this paper USES a camera and a projector, the projector projects the encoded sinusoidal grating image to the measured object, and the camera collects the images of the measured object illuminated by the grating. According to the three-step phase shift method, the relative phase of the projected grating can be obtained. After phase unrolling, its absolute phase can be obtained. However, for the sinusoidal grating in the row direction, its absolute phase is unique on any row, that is, the absolute phase value can be used to uniquely determine a pixel point. Then, we use the sine grating in the column direction to solve the absolute phase in the column direction, so that the absolute phase in the row direction and the absolute phase in the column direction can be uniquely determined. Therefore, point matching can be realized by solving the absolute phase of two images.

Image points matching method²⁰ based on absolute phase constraint is based on the projector and camera images of each point in the absolute phase values to match, the image noise points, because itself exists to some pretreatment, so won't be affected by its grey value itself, the matching method based on phase and matching method based on the similarity between the different nature, because of the phase out is obtained according to the three phase shift method, while the input is the gray value of each grating, but the output phase and only three grating of gray level difference, and neither one image grey value is directly related to Compared with the gray similarity matching method, the absolute phase matching is more accurate and reliable. Obviously, when point matching is performed with and using absolute phase, the accuracy of matching is determined by the solution accuracy of absolute phase, so this method depends on the phase solution of grating image and deformed grating image and the precision of phase unwrapping of wrapping phase.

6. The experimental results

6.1. Raster image

The experimental unwrapping phase diagrams of left camera, right camera and 3D point cloud images are shown in the following figure:



Fig. 4 (a) Left camera unwrapping image; (b)Right camera unwrap image; (c)Cloud image

6.2. The error analysis

According to the experimental data, the dimensions of the above objects were measured. The test results are as follows:

Table 1. Comparison of test data with error data.

measurement	The actual size /mm	The measured results /mm	error /mm
1	2.0533	2.1248	0.0715
2	3.8991	3.9409	0.0418
3	7.0911	7.1716	0.0805
4	5.0974	5.1449	0.0475
5	5.8983	5.8334	-0.0649

7. Conclusion

Based on the principle of binocular vision and grating projection measurement, a 3d measurement system is built to verify the feasibility of the system. The three-dimensional point cloud of the part is obtained by measuring the object, which indicates that the system can realize the three-dimensional size measurement of the complex surface topography and has high measurement efficiency. According to the measurement results, the measurement accuracy of this system is 0.1mm, which is expected to replace the existing manual measurement method. Meanwhile, the research work in this paper can provide reference and reference for the 3D detection of complex polyhedron parts and large structural parts in the future.

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